УО «Белорусский государственный университет информатики и радиоэлектроники»
Кафедра ПОИТ

Отчет по лабораторной работе №2
по предмету
Распределенные вычисления
Вариант 1

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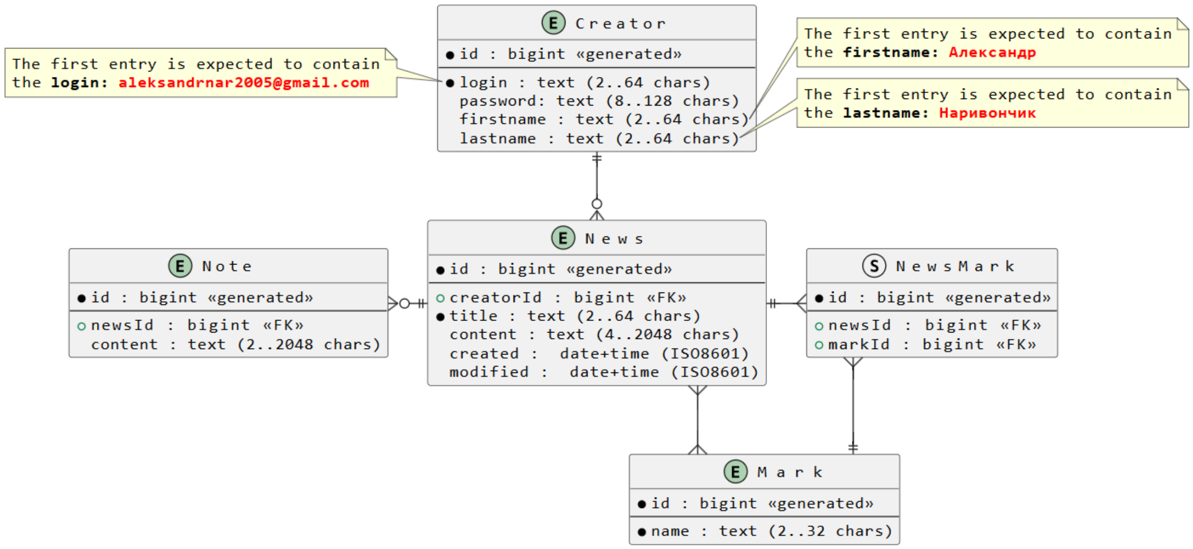
Минск 2026

Задание

Внимание, префикс таблиц не указан, но обязательно должен быть в решении



- Модель данных:
  - Используются сущности из предыдущей работы, указанные на схеме
    - **Creator**
    - **News**
    - **Mark**
    - **Note**
    - при необходимости можно использовать общую абстракцию для сущностей
- Слой хранения:
  - Составьте сценарий инициализации базы данных, рекомендуется использовать **liquibase** в формате XML
  - Проверьте создание таблиц **tbl_creator**, **tbl_news**, **tbl_mark** и **tbl_note**.
  - Доработайте обобщенный интерфейс для хранения и поиска данных сущностей для **CRUD** операций:
    - операции поиска и выборки должны поддерживать пагинацию, фильтрацию и сортировку
  - Протестируйте разработанный интерфейс для сущностей **Creator**, **News**, **Mark** и **Note**.

Технические требования

- **ВАЖНО!** Используйте префикс **/api/v1.0/** для контроллеров REST и их методов,

- **ВАЖНО!** Используйте адрес и порт **localhost:24110** для самого приложения.
- **ВАЖНО!** Используйте следующие данные при подключении к **Postgresql**
  - Префикс драйвера **jdbc:postgresql:**
  - Адрес базы данных **localhost**
  - Порт базы данных **5432**
  - user **postgres**
  - password **postgres**
  - Схема **distcomp**
- **ВАЖНО!** Используйте обязательный префикс **tbl_** для таблиц в базе данных

Код

```
using Microsoft.AspNetCore.Mvc;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.application.exceptions;
using rest1.application.interfaces.services;

namespace rest1.api.controllers;

    [ApiController]
    [Route("api/v1.0/[controller]")]
    public class CreatorsController : ControllerBase
    {
        private readonly ICreatorService _creatorService;
        private readonly ILogger<CreatorsController> _logger;

        public CreatorsController(ICreatorService creatorService,
ILogger<CreatorsController> logger)
        {
            _creatorService = creatorService;
            _logger = logger;
        }

        [HttpPost]
        [ProducesResponseType(typeof(CreatorResponseTo),
```

```
StatusCodes.Status201Created)]
        [ProducesResponseType(StatusCodes.Status400BadRequest)]
        [ProducesResponseType(StatusCodes.Status500InternalServerError)]
        public async Task<ActionResult<CreatorResponseTo>> CreateCreator(
            [FromBody] CreatorRequestTo createCreatorRequest)
        {
            try
            {
                _logger.LogInformation("Creating creator {request}",
createCreatorRequest);

                CreatorResponseTo createdCreator =
                    await _creatorService.CreateCreator(createCreatorRequest);

                return CreatedAtAction(
                    nameof(CreateCreator),
                    new { id = createdCreator.Id },
                    createdCreator
                );
            }
            catch (Exception ex)
            {
                _logger.LogError(ex, "Error creating creator");
                return StatusCode(500, "An error occurred while creating the creator");
            }
        }

        [HttpGet]
        [ProducesResponseType(typeof(IEnumerable<CreatorResponseTo>),
StatusCodes.Status200OK)]
        [ProducesResponseType(StatusCodes.Status500InternalServerError)]
        public async Task<ActionResult<IEnumerable<CreatorResponseTo>>>
GetAllCreators()
        {
            try
            {
                _logger.LogInformation("Getting all creators");

                IEnumerable<CreatorResponseTo> creators =
                    await _creatorService.GetAllCreators();
```

```
            return Ok(creators);
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error getting all creators");
            return StatusCode(500, "An error occurred while retrieving creators");
        }
    }

    [HttpGet("{id:long}")]
    [ProducesResponseType(typeof(CreatorResponseTo),
StatusCodes.Status200OK)]
    [ProducesResponseType(StatusCodes.Status404NotFound)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<CreatorResponseTo>>
GetCreatorById([FromRoute] long id)
    {
        try
        {
            _logger.LogInformation("Getting creator by id: {Id}", id);

            var getCreatorRequest = new CreatorRequestTo { Id = id };
            CreatorResponseTo creator =
                await _creatorService.GetCreator(getCreatorRequest);

            return Ok(creator);
        }
        catch (NewNotFoundException ex)
        {
            _logger.LogWarning(ex, "Creator not found with id: {Id}", id);
            return NotFound($"Creator with id {id} not found");
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error getting creator by id: {Id}", id);
            return StatusCode(500, "An error occurred while retrieving the creator");
        }
    }
```

```
        [HttpPut]
        [ProducesResponseType(typeof(CreatorResponseTo),
    StatusCodes.Status200OK)]
        [ProducesResponseType(StatusCodes.Status400BadRequest)]
        [ProducesResponseType(StatusCodes.Status404NotFound)]
        [ProducesResponseType(StatusCodes.Status500InternalServerError)]
        public async Task<ActionResult<CreatorResponseTo>> UpdateCreator(
            [FromBody] CreatorRequestTo updateCreatorRequest)
        {
            try
            {
                _logger.LogInformation(
                    "Updating creator with id: {Id}",
                    updateCreatorRequest.Id);

                var updatedCreator =
                    await _creatorService.UpdateCreator(updateCreatorRequest);

                return Ok(updatedCreator);
            }
            catch (NewNotFoundException ex)
            {
                _logger.LogWarning(
                    ex,
                    "Creator not found for update with id: {Id}",
                    updateCreatorRequest.Id);

                return NotFound(
                    $"Creator with id {updateCreatorRequest.Id} not found");
            }
            catch (Exception ex)
            {
                _logger.LogError(
                    ex,
                    "Error updating creator with id: {Id}",
                    updateCreatorRequest.Id);

                return StatusCode(
                    500,
                    "An error occurred while updating the creator");
```

```
        }
    }

    [HttpDelete("{id:long}")]
    [ProducesResponseType(StatusCodes.Status204NoContent)]
    [ProducesResponseType(StatusCodes.Status404NotFound)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<IActionResult> DeleteCreator([FromRoute] long id)
    {
        try
        {
            _logger.LogInformation("Deleting creator with id: {Id}", id);

            var deleteCreatorRequest = new CreatorRequestTo { Id = id };
            await _creatorService.DeleteCreator(deleteCreatorRequest);

            return NoContent();
        }
        catch (CreatorNotFoundException ex)
        {
            _logger.LogWarning(
                ex,
                "Creator not found for deletion with id: {Id}",
                id);

            return NotFound();
        }
        catch (Exception ex)
        {
            _logger.LogError(
                ex,
                "Error deleting creator with id: {Id}",
                id);

            return StatusCode(
                500,
                "An error occurred while deleting the creator");
        }
    }
}
```

```
using Microsoft.AspNetCore.Mvc;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.application.exceptions;
using rest1.application.interfaces.services;

namespace rest1.api.controllers;

[ApiController]
[Route("api/v1.0/[controller]")]
public class MarksController : ControllerBase
{
    private readonly IMarkService _markService;
    private readonly ILogger<MarksController> _logger;

    public MarksController(IMarkService markService, ILogger<MarksController>
logger)
    {
        _markService = markService;
        _logger = logger;
    }

    [HttpPost]
    [ProducesResponseType(typeof(MarkResponseTo),
StatusCodes.Status201Created)]
    [ProducesResponseType(StatusCodes.Status400BadRequest)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<MarkResponseTo>> CreateMark(
        [FromBody] MarkRequestTo createMarkRequest)
    {
        try
        {
            _logger.LogInformation("Creating mark {request}", createMarkRequest);

            MarkResponseTo createdMark =
                await _markService.CreateMark(createMarkRequest);

            return CreatedAtAction(
                nameof(CreateMark),
```

```
            new { id = createdMark.Id },
            createdMark
        );
    }
    catch (Exception ex)
    {
        _logger.LogError(ex, "Error creating mark");
        return StatusCode(500, "An error occurred while creating the mark");
    }
}

[HttpGet]
[ProducesResponseType(typeof(IEnumerable<MarkResponseTo>),
StatusCodes.Status200OK)]
[ProducesResponseType(StatusCodes.Status500InternalServerError)]
public async Task<ActionResult<IEnumerable<MarkResponseTo>>>
GetAllMarks()
{
    try
    {
        _logger.LogInformation("Getting all marks");

        IEnumerable<MarkResponseTo> marks =
            await _markService.GetAllMarks();

        return Ok(marks);
    }
    catch (Exception ex)
    {
        _logger.LogError(ex, "Error getting all marks");
        return StatusCode(500, "An error occurred while retrieving marks");
    }
}

[HttpGet("{id:long}")]
[ProducesResponseType(typeof(MarkResponseTo), StatusCodes.Status200OK)]
[ProducesResponseType(StatusCodes.Status404NotFound)]
[ProducesResponseType(StatusCodes.Status500InternalServerError)]
public async Task<ActionResult<MarkResponseTo>>
GetMarkById([FromRoute] long id)
```

```
    {
        try
        {
            _logger.LogInformation("Getting mark by id: {Id}", id);

            var getMarkRequest = new MarkRequestTo { Id = id };
            MarkResponseTo mark =
                await _markService.GetMark(getMarkRequest);

            return Ok(mark);
        }
        catch (NewNotFoundException ex)
        {
            _logger.LogWarning(ex, "Mark not found with id: {Id}", id);
            return NotFound($"Mark with id {id} not found");
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error getting mark by id: {Id}", id);
            return StatusCode(500, "An error occurred while retrieving the mark");
        }
    }

    [HttpPut]
    [ProducesResponseType(typeof(MarkResponseTo), StatusCodes.Status200OK)]
    [ProducesResponseType(StatusCodes.Status400BadRequest)]
    [ProducesResponseType(StatusCodes.Status404NotFound)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<MarkResponseTo>> UpdateMark(
        [FromBody] MarkRequestTo updateMarkRequest)
    {
        try
        {
            _logger.LogInformation("Updating mark with id: {Id}",
updateMarkRequest.Id);

            var updatedMark =
                await _markService.UpdateMark(updateMarkRequest);

            return Ok(updatedMark);
```

```
    }
    catch (NewNotFoundException ex)
    {
        _logger.LogWarning(
            ex,
            "Mark not found for update with id: {Id}",
            updateMarkRequest.Id);

        return NotFound(
            $"Mark with id {updateMarkRequest.Id} not found");
    }
    catch (Exception ex)
    {
        _logger.LogError(
            ex,
            "Error updating mark with id: {Id}",
            updateMarkRequest.Id);

        return StatusCode(
            500,
            "An error occurred while updating the mark");
    }
}

[HttpDelete("{id:long}")]
[ProducesResponseType(StatusCodes.Status204NoContent)]
[ProducesResponseType(StatusCodes.Status404NotFound)]
[ProducesResponseType(StatusCodes.Status500InternalServerError)]
public async Task<IActionResult> DeleteMark([FromRoute] long id)
{
    try
    {
        _logger.LogInformation("Deleting mark with id: {Id}", id);

        var deleteMarkRequest = new MarkRequestTo { Id = id };
        await _markService.DeleteMark(deleteMarkRequest);

        return NoContent();
    }
    catch (MarkNotFoundException ex)
```

```
        {
            _logger.LogWarning(ex, "Mark not found for deletion with id: {Id}", id);
            return NotFound();
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error deleting mark with id: {Id}", id);
            return StatusCode(500, "An error occurred while deleting the mark");
        }
    }
}

using Microsoft.AspNetCore.Mvc;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.application.exceptions;
using rest1.application.interfaces.services;

namespace rest1.api.controllers;

[ApiController]
[Route("api/v1.0/[controller]")]
public class NewsController : ControllerBase
{
    private readonly INewsService _newsService;
    private readonly ILogger<NewsController> _logger;

    public NewsController(INewsService newsService, ILogger<NewsController>
logger)
    {
        _newsService = newsService;
        _logger = logger;
    }

    [HttpPost]
    [ProducesResponseType(typeof(NewsResponseTo),
StatusCodes.Status201Created)]
    [ProducesResponseType(StatusCodes.Status400BadRequest)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<NewsResponseTo>>
```

```
CreateNews([FromBody] NewsRequestTo createNewsRequest)
    {
        try
        {
            _logger.LogInformation("Creating news with title: {Title}",
createNewsRequest.Title);

            var createdNews = await _newsService.CreateNews(createNewsRequest);

            return CreatedAtAction(
                nameof(GetNewsById),
                new { id = createdNews.Id },
                createdNews
            );
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error creating news");
            return StatusCode(500, "An error occurred while creating the news");
        }
    }

    [HttpGet]
    [ProducesResponseType(typeof(IEnumerable<NewsResponseTo>),
StatusCodes.Status200OK)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<IEnumerable<NewsResponseTo>>>
GetAllNews()
    {
        try
        {
            _logger.LogInformation("Getting all news");

            var news = await _newsService.GetAllNews();

            return Ok(news);
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error getting all news");
```

```
            return StatusCode(500, "An error occurred while retrieving news");
        }
    }

    [HttpGet("{id:long}")]
    [ProducesResponseType(typeof(NewsResponseTo), StatusCodes.Status200OK)]
    [ProducesResponseType(StatusCodes.Status404NotFound)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<NewsResponseTo>>
GetNewsById([FromRoute] long id)
    {
        try
        {
            _logger.LogInformation("Getting news by id: {Id}", id);

            var getNewsRequest = new NewsRequestTo { Id = id };
            var news = await _newsService.GetNews(getNewsRequest);

            return Ok(news);
        }
        catch (NewNotFoundException ex)
        {
            _logger.LogWarning(ex, "News not found with id: {Id}", id);
            return NotFound($"News with id {id} not found");
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error getting news by id: {Id}", id);
            return StatusCode(500, "An error occurred while retrieving the news");
        }
    }

    [HttpPut]
    [ProducesResponseType(typeof(NewsResponseTo), StatusCodes.Status200OK)]
    [ProducesResponseType(StatusCodes.Status400BadRequest)]
    [ProducesResponseType(StatusCodes.Status404NotFound)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<NewsResponseTo>> UpdateNews(
        [FromBody] NewsRequestTo updateNewsRequest)
    {
```

```
        try
        {
            _logger.LogInformation("Updating news with id: {Id}",
updateNewsRequest.Id);

            var updatedNews = await
_newsService.UpdateNews(updateNewsRequest);

            return Ok(updatedNews);
        }
        catch (NewNotFoundException ex)
        {
            _logger.LogWarning(ex, "News not found for update with id: {Id}",
updateNewsRequest.Id);
            return NotFound();
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error updating news with id: {Id}",
updateNewsRequest.Id);
            return StatusCode(500, "An error occurred while updating the news");
        }
    }

    [HttpDelete("{id:long}")]
    [ProducesResponseType(StatusCodes.Status204NoContent)]
    [ProducesResponseType(StatusCodes.Status404NotFound)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<IActionResult> DeleteNews([FromRoute] long id)
    {
        try
        {
            _logger.LogInformation("Deleting news with id: {Id}", id);

            var deleteNewsRequest = new NewsRequestTo { Id = id };
            await _newsService.DeleteNews(deleteNewsRequest);

            return NoContent();
        }
        catch (NewNotFoundException ex)
```

```
        {
            _logger.LogWarning(ex, "News not found for deletion with id: {Id}", id);
            return NotFound();
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error deleting news with id: {Id}", id);
            return StatusCode(500, "An error occurred while deleting the news");
        }
    }
}

using Microsoft.AspNetCore.Mvc;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.application.exceptions;
using rest1.application.interfaces.services;

namespace rest1.api.controllers;

    [ApiController]
    [Route("api/v1.0/[controller]")]
    public class NotesController : ControllerBase
    {
        private readonly INoteService _noteService;
        private readonly ILogger<NotesController> _logger;

        public NotesController(INoteService noteService, ILogger<NotesController>
logger)
        {
            _noteService = noteService;
            _logger = logger;
        }

        [HttpPost]
        [ProducesResponseType(typeof(NoteResponseTo),
StatusCodes.Status201Created)]
        [ProducesResponseType(StatusCodes.Status400BadRequest)]
        [ProducesResponseType(StatusCodes.Status500InternalServerError)]
        public async Task<ActionResult<NoteResponseTo>> CreateNote(
```

```
            [FromBody] NoteRequestTo createNoteRequest)
        {
            try
            {
                _logger.LogInformation("Creating note {request}", createNoteRequest);

                NoteResponseTo createdNote =
                    await _noteService.CreateNote(createNoteRequest);

                return CreatedAtAction(
                    nameof(CreateNote),
                    new { id = createdNote.Id },
                    createdNote
                );
            }
            catch (Exception ex)
            {
                _logger.LogError(ex, "Error creating note");
                return StatusCode(500, "An error occurred while creating the note");
            }
        }

        [HttpGet]
        [ProducesResponseType(typeof(IEnumerable<NoteResponseTo>),
StatusCodes.Status200OK)]
        [ProducesResponseType(StatusCodes.Status500InternalServerError)]
        public async Task<ActionResult<IEnumerable<NoteResponseTo>>>
GetAllNotes()
        {
            try
            {
                _logger.LogInformation("Getting all notes");

                IEnumerable<NoteResponseTo> notes =
                    await _noteService.GetAllNotes();

                return Ok(notes);
            }
            catch (Exception ex)
            {
```

```
            _logger.LogError(ex, "Error getting all notes");
            return StatusCode(500, "An error occurred while retrieving notes");
        }
    }

    [HttpGet("{id:long}")]
    [ProducesResponseType(typeof(NoteResponseTo),
StatusCodes.Status200OK)]
    [ProducesResponseType(StatusCodes.Status404NotFound)]
    [ProducesResponseType(StatusCodes.Status500InternalServerError)]
    public async Task<ActionResult<NoteResponseTo>>
GetNoteById([FromRoute] long id)
    {
        try
        {
            _logger.LogInformation("Getting note by id: {Id}", id);

            var getNoteRequest = new NoteRequestTo { Id = id };
            NoteResponseTo note =
                await _noteService.GetNote(getNoteRequest);

            return Ok(note);
        }
        catch (NewNotFoundException ex)
        {
            _logger.LogWarning(ex, "Note not found with id: {Id}", id);
            return NotFound($"Note with id {id} not found");
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Error getting note by id: {Id}", id);
            return StatusCode(500, "An error occurred while retrieving the note");
        }
    }

    [HttpPut]
    [ProducesResponseType(typeof(NoteResponseTo),
StatusCodes.Status200OK)]
    [ProducesResponseType(StatusCodes.Status400BadRequest)]
    [ProducesResponseType(StatusCodes.Status404NotFound)]
```

```
        [ProducesResponseType(StatusCodes.Status500InternalServerError)]
        public async Task<ActionResult<NoteResponseTo>> UpdateNote(
            [FromBody] NoteRequestTo updateNoteRequest)
        {
            try
            {
                _logger.LogInformation("Updating note with id: {Id}",
updateNoteRequest.Id);

                var updatedNote =
                    await _noteService.UpdateNote(updateNoteRequest);

                return Ok(updatedNote);
            }
            catch (NewNotFoundException ex)
            {
                _logger.LogWarning(
                    ex,
                    "Note not found for update with id: {Id}",
                    updateNoteRequest.Id);

                return NotFound(
                    $"Note with id {updateNoteRequest.Id} not found");
            }
            catch (Exception ex)
            {
                _logger.LogError(
                    ex,
                    "Error updating note with id: {Id}",
                    updateNoteRequest.Id);

                return StatusCode(
                    500,
                    "An error occurred while updating the note");
            }
        }

        [HttpDelete("{id:long}")]
        [ProducesResponseType(StatusCodes.Status204NoContent)]
        [ProducesResponseType(StatusCodes.Status404NotFound)]
```

```
        [ProducesResponseType(StatusCodes.Status500InternalServerError)]
        public async Task<IActionResult> DeleteNote([FromRoute] long id)
        {
            try
            {
                _logger.LogInformation("Deleting note with id: {Id}", id);

                var deleteNoteRequest = new NoteRequestTo { Id = id };
                await _noteService.DeleteNote(deleteNoteRequest);

                return NoContent();
            }
            catch (NoteNotFoundException ex)
            {
                _logger.LogWarning(ex, "Note not found for deletion with id: {Id}", id);
                return NotFound();
            }
            catch (Exception ex)
            {
                _logger.LogError(ex, "Error deleting note with id: {Id}", id);
                return StatusCode(500, "An error occurred while deleting the note");
            }
        }
    }

using System.ComponentModel.DataAnnotations;

namespace rest1.application.DTOs.requests;

public class CreatorRequestTo
{
        public long? Id { get; set; }

        [StringLength(64, MinimumLength = 2)]
        public string? Login { get; set; }

        [StringLength(128, MinimumLength = 8)]
        public string? Password { get; set; }

        [StringLength(64, MinimumLength = 2)]
```

```
    public string? Firstname { get; set; }

    [StringLength(64, MinimumLength = 2)]
    public string? Lastname { get; set; }
}

using System.ComponentModel.DataAnnotations;

namespace rest1.application.DTOs.requests;

public class MarkRequestTo
{
  public long? Id { get; set; }

  [StringLength(32, MinimumLength = 2)]
  public string? Name { get; set; }

  public long[] NewsIds { get; set; }
}

using System.ComponentModel.DataAnnotations;

namespace rest1.application.DTOs.requests;

public class NewsRequestTo
{
  public long? Id { get; set; }

  public long CreatorId { get; set; }

  [StringLength(64, MinimumLength = 2)]
  public string Title { get; set; } = string.Empty;

  [StringLength(2048, MinimumLength = 4)]
  public string Content { get; set; } = string.Empty;

  public long[] MarkIds { get; set; }
}

using System.ComponentModel.DataAnnotations;
```

```
namespace rest1.application.DTOs.requests;

public class NoteRequestTo
{
    public long? Id { get; set; }

    public long NewsId { get; set; }

    [StringLength(2048, MinimumLength = 2)]
    public string Content { get; set; }
}

namespace rest1.application.DTOs.responses;

public class CreatorResponseTo (
    string login,
    string password,
    string firstName,
    string lastName)
{
    public long Id { get; set; }

    public string Login { get; set; } = login;

    public string Password { get; set; } = password;

    public string Firstname { get; set; } = firstName;

    public string Lastname { get; set; } = lastName;

}

namespace rest1.application.DTOs.responses;

public class MarkResponseTo(
        long id,
        string name)
{
    public long? Id { get; set; } = id;
```

```
    public string? Name { get; set; } = name;
}

namespace rest1.application.DTOs.responses;

public class NewsResponseTo(
    string title,
    string content,
    DateTime createdAt,
    DateTime modified)
{
    public long Id { get; set; }

    public long CreatorId { get; set; }

    public string Title { get; set; } = title;

    public string Content { get; set; } = content;

    public DateTime CreatedAt { get; set; } = createdAt;

    public DateTime Modified { get; set; } = modified;
}

namespace rest1.application.DTOs.responses;

public class NoteResponseTo(
        long id,
        long newsId,
        string content)
{
    public long Id { get; set; } = id;

    public long NewsId { get; set; } = newsId;

    public string Content {  get; set; } = content;
}

namespace rest1.application.exceptions;
```

```
public class CreatorNotFoundException : NotFoundException
{
   public CreatorNotFoundException(string message, Exception inner) :
base(message, inner) { }

   public CreatorNotFoundException(string message) : base(message) { }
}

namespace rest1.application.exceptions;

public class MarkNotFoundException : NotFoundException
{
   public MarkNotFoundException(string message, Exception inner) :
base(message, inner) { }

   public MarkNotFoundException(string message) : base(message) { }
}

namespace rest1.application.exceptions;

public class NewNotFoundException : NotFoundException
{
   public NewNotFoundException(string message, Exception inner) :
base(message, inner) { }

   public NewNotFoundException(string message) : base(message) { }
}

namespace rest1.application.exceptions;

public class NotFoundException : Exception
{
   public NotFoundException(string message, Exception inner) : base(message,
inner) { }

   public NotFoundException(string message) : base(message) { }
}

namespace rest1.application.exceptions;
```

```
public class NoteNotFoundException : NotFoundException
{
    public NoteNotFoundException(string message, Exception inner) :
base(message, inner) { }

    public NoteNotFoundException(string message) : base(message) { }
}

using rest1.core.entities;

namespace rest1.application.interfaces;

public interface ICreatorRepository : IRepository<Creator>
{

}

using rest1.core.entities;

namespace rest1.application.interfaces;

public interface IMarkRepository : IRepository<Mark>
{

}

using rest1.core.entities;

namespace rest1.application.interfaces;

public interface INewsRepository : IRepository<News>
{

}

using rest1.core.entities;

namespace rest1.application.interfaces;
```

```
public interface INoteRepository : IRepository<Note>
{

}

using rest1.core.entities;

namespace rest1.application.interfaces;

public interface IRepository<TEntity> where TEntity : Entity
{
   //get records
   Task<TEntity?> GetByIdAsync(long id, CancellationToken cancellationToken
= default);
   Task<IEnumerable<TEntity>> GetAllAsync(CancellationToken ct = default);

   //create record
   Task<TEntity> AddAsync(TEntity entity, CancellationToken ct = default);

   //update record
   Task<TEntity?> UpdateAsync(TEntity entity, CancellationToken ct = default);

   //delete record
   Task<TEntity?> DeleteAsync(TEntity entity, CancellationToken ct = default);

   //find records
   Task<TEntity?> FindAsync(ISpecification<TEntity> spec, CancellationToken
ct = default);
   Task<IEnumerable<TEntity>> FindAllAsync(ISpecification<TEntity> spec,
CancellationToken ct = default);

   //count records
   Task<int> CountAsync(ISpecification<TEntity> spec, CancellationToken ct =
default);

   //is any record
   Task<bool> AnyAsync(ISpecification<TEntity> spec, CancellationToken ct =
default);
}
```

```
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;

namespace rest1.application.interfaces.services;

public interface ICreatorService
{
    Task<CreatorResponseTo> CreateCreator(CreatorRequestTo
createCreatorRequestTo);

    Task<IEnumerable<CreatorResponseTo>> GetAllCreators();

    Task<CreatorResponseTo> GetCreator(CreatorRequestTo
getCreatorRequestTo);

    Task<CreatorResponseTo> UpdateCreator(CreatorRequestTo
updateCreatorRequestTo);

    Task DeleteCreator(CreatorRequestTo deleteCreatorRequestTo);
}

using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;

namespace rest1.application.interfaces.services;

public interface IMarkService
{
    Task<MarkResponseTo> CreateMark(MarkRequestTo createMarkRequestTo);

    Task<IEnumerable<MarkResponseTo>> GetAllMarks();

    Task<MarkResponseTo> GetMark(MarkRequestTo getMarkRequestTo);

    Task<MarkResponseTo> UpdateMark(MarkRequestTo updateMarkRequestTo);

    Task DeleteMark(MarkRequestTo deleteMarkRequestTo);
}

using rest1.application.DTOs.requests;
```

```
using rest1.application.DTOs.responses;

namespace rest1.application.interfaces.services;

public interface INewsService
{
    Task<NewsResponseTo> CreateNews(NewsRequestTo createNewsRequestTo);

    Task<IEnumerable<NewsResponseTo>> GetAllNews();

    Task<NewsResponseTo> GetNews(NewsRequestTo getNewsRequestTo);

    Task<NewsResponseTo> UpdateNews(NewsRequestTo
updateNewsRequestTo);

    Task DeleteNews(NewsRequestTo deleteNewsRequestTo);
}

using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;

namespace rest1.application.interfaces.services;

public interface INoteService
{
    Task<NoteResponseTo> CreateNote(NoteRequestTo createNoteRequestTo);

    Task<IEnumerable<NoteResponseTo>> GetAllNotes();

    Task<NoteResponseTo> GetNote(NoteRequestTo getNoteRequestTo);

    Task<NoteResponseTo> UpdateNote(NoteRequestTo updateNoteRequestTo);

    Task DeleteNote(NoteRequestTo deleteNoteRequestTo);
}

using System.Linq.Expressions;

namespace rest1.application.interfaces;
```

```
public interface ISpecification<T>
{
    //criteria expression
    Expression<Func<T, bool>> Criteria { get; }

    //join expressions
    List<Expression<Func<T, object>>> Includes { get; }

    //order by expressions
    Expression<Func<T, object>>? OrderBy { get; }
    Expression<Func<T, object>>? OrderByDescending { get; }

    //taken records count
    int? Take { get; }

    //skipped records count
    int? Skip { get; }

    //pagination flag
    bool IsPagingEnabled { get; }
}
```

```
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.core.entities;
using AutoMapper;

namespace rest1.application.mappers;

public class CreatorProfile : Profile
{
    public CreatorProfile()
    {
        CreateMap<CreatorRequestTo, Creator>()
            .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id ?? 0))
            .ForMember(dest => dest.Login, opt => opt.MapFrom(src => src.Login))
            .ForMember(dest => dest.Firstname, opt => opt.MapFrom(src =>
src.Firstname))
            .ForMember(dest => dest.Lastname, opt => opt.MapFrom(src =>
src.Lastname))
```

```
            .ForMember(dest => dest.Password, opt => opt.MapFrom(src =>
src.Password));

        CreateMap<Creator, CreatorResponseTo>()
            .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id))
            .ForMember(dest => dest.Login, opt => opt.MapFrom(src => src.Login))
            .ForMember(dest => dest.Firstname, opt => opt.MapFrom(src =>
src.Firstname))
            .ForMember(dest => dest.Lastname, opt => opt.MapFrom(src =>
src.Lastname))
            .ForMember(dest => dest.Password, opt => opt.MapFrom(src =>
src.Password));
    }
}

using AutoMapper;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.core.entities;

namespace rest1.application.mappers;

public class MarkProfile : Profile
{
    public MarkProfile()
    {
        CreateMap<MarkRequestTo, Mark>()
            .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id ?? 0))
            .ForMember(dest => dest.Name, opt => opt.MapFrom(src => src.Name));

        CreateMap<Mark, MarkResponseTo>()
            .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id))
            .ForMember(dest => dest.Name, opt => opt.MapFrom(src => src.Name));
    }
}

using AutoMapper;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.core.entities;
```

```
namespace rest1.application.mappers;

public class NewsProfile : Profile
{
    public NewsProfile()
    {
        CreateMap<NewsRequestTo, News>()
            .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id ?? 0))
            .ForMember(dest => dest.CreatedAt, opt => opt.MapFrom(src =>
DateTime.UtcNow))
            .ForMember(dest => dest.Modified, opt => opt.MapFrom(src =>
DateTime.UtcNow))
            .ForMember(dest => dest.CreatorId, opt => opt.MapFrom(src =>
src.CreatorId))
            .ForMember(dest => dest.Content, opt => opt.MapFrom(src =>
src.Content))
            .ForMember(dest => dest.Title, opt => opt.MapFrom(src => src.Title));

        CreateMap<News, NewsResponseTo>()
            .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id))
            .ForMember(dest => dest.CreatorId, opt => opt.MapFrom(src =>
src.CreatorId))
            .ForMember(dest => dest.Title, opt => opt.MapFrom(src => src.Title))
            .ForMember(dest => dest.Content, opt => opt.MapFrom(src =>
src.Content))
            .ForMember(dest => dest.CreatedAt, opt => opt.MapFrom(src =>
src.CreatedAt))
            .ForMember(dest => dest.Modified, opt => opt.MapFrom(src =>
src.Modified));
    }
}

using AutoMapper;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.core.entities;

namespace rest1.application.mappers;
```

```
public class NoteProfile : Profile
{
    public NoteProfile()
    {
        CreateMap<NoteRequestTo, Note>()
            .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id ?? 0))
            .ForMember(dest => dest.NewsId, opt => opt.MapFrom(src =>
src.NewsId));

        CreateMap<Note, NoteResponseTo>()
            .ForMember(dest => dest.Id, opt => opt.MapFrom(src => src.Id))
            .ForMember(dest => dest.NewsId, opt => opt.MapFrom(src =>
src.NewsId));
    }
}

using AutoMapper;
using Microsoft.AspNetCore.Components.Forms;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.application.exceptions;
using rest1.application.interfaces;
using rest1.application.interfaces.services;
using rest1.core.entities;

namespace rest1.application.services;

public class CreatorService : ICreatorService
{
    private readonly IMapper _mapper;
    private readonly ICreatorRepository _creatorRepository;

    public CreatorService(IMapper mapper, ICreatorRepository repository)
    {
        _mapper = mapper;
        _creatorRepository = repository;
    }

    public async Task<CreatorResponseTo> CreateCreator(CreatorRequestTo
createCreatorRequestTo)
```

```
    {
        Creator creatorFromDto = _mapper.Map<Creator>(createCreatorRequestTo);

        Creator createdCreator = await
_creatorRepository.AddAsync(creatorFromDto);

        CreatorResponseTo dtoFromCreatedEditor =
_mapper.Map<CreatorResponseTo>(createdCreator);

        return dtoFromCreatedEditor;
    }

    public async Task DeleteCreator(CreatorRequestTo deleteCreatorRequestTo)
    {
        Creator creatorFromDto = _mapper.Map<Creator>(deleteCreatorRequestTo);

        _ = await _creatorRepository.DeleteAsync(creatorFromDto)
            ?? throw new CreatorNotFoundException(
                $"Delete creator {creatorFromDto} was not found");
    }

    public async Task<IEnumerable<CreatorResponseTo>> GetAllCreators()
    {
        IEnumerable<Creator> allCreators =
            await _creatorRepository.GetAllAsync();

        var response = new List<CreatorResponseTo>();

        foreach (Creator creator in allCreators)
        {
            response.Add(_mapper.Map<CreatorResponseTo>(creator));
        }

        return response;
    }

    public async Task<CreatorResponseTo> GetCreator(CreatorRequestTo
getCreatorRequestTo)
    {
        Creator creatorFromDto = _mapper.Map<Creator>(getCreatorRequestTo);
```

```
        Creator demandedCreator =
            await _creatorRepository.GetByIdAsync(creatorFromDto.Id)
            ?? throw new CreatorNotFoundException(
                $"Demanded creator {creatorFromDto} was not found");

        return _mapper.Map<CreatorResponseTo>(demandedCreator);
    }

    public async Task<CreatorResponseTo> UpdateCreator(CreatorRequestTo
updateCreatorRequestTo)
    {
        Creator creatorFromDto = _mapper.Map<Creator>(updateCreatorRequestTo);

        Creator updatedCreator =
            await _creatorRepository.UpdateAsync(creatorFromDto)
            ?? throw new CreatorNotFoundException(
                $"Update creator {creatorFromDto} was not found");

        return _mapper.Map<CreatorResponseTo>(updatedCreator);
    }
}

using AutoMapper;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.application.exceptions;
using rest1.application.interfaces;
using rest1.application.interfaces.services;
using rest1.core.entities;

namespace rest1.application.services;

public class MarkService : IMarkService
{
    private readonly IMapper _mapper;
    private readonly IMarkRepository _markRepository;

    public MarkService(IMapper mapper, IMarkRepository repository)
    {
```

```
        _mapper = mapper;
        _markRepository = repository;
    }

    public async Task<MarkResponseTo> CreateMark(MarkRequestTo
createMarkRequestTo)
    {
        Mark markFromDto = _mapper.Map<Mark>(createMarkRequestTo);

        Mark createdMark = await _markRepository.AddAsync(markFromDto);

        MarkResponseTo dtoFromCreatedMark =
            _mapper.Map<MarkResponseTo>(createdMark);

        return dtoFromCreatedMark;
    }

    public async Task DeleteMark(MarkRequestTo deleteMarkRequestTo)
    {
        Mark markFromDto = _mapper.Map<Mark>(deleteMarkRequestTo);

        _ = await _markRepository.DeleteAsync(markFromDto)
            ?? throw new MarkNotFoundException(
                $"Delete mark {markFromDto} was not found");
    }

    public async Task<IEnumerable<MarkResponseTo>> GetAllMarks()
    {
        IEnumerable<Mark> allMarks =
            await _markRepository.GetAllAsync();

        var allMarksResponseTos = new List<MarkResponseTo>();

        foreach (Mark mark in allMarks)
        {
            MarkResponseTo markTo =
                _mapper.Map<MarkResponseTo>(mark);

            allMarksResponseTos.Add(markTo);
        }
```

```
        return allMarksResponseTos;
    }

    public async Task<MarkResponseTo> GetMark(MarkRequestTo
getMarksRequestTo)
    {
        Mark markFromDto = _mapper.Map<Mark>(getMarksRequestTo);

        Mark demandedMark =
            await _markRepository.GetByIdAsync(markFromDto.Id)
            ?? throw new MarkNotFoundException(
                $"Demanded mark {markFromDto} was not found");

        MarkResponseTo demandedMarkResponseTo =
            _mapper.Map<MarkResponseTo>(demandedMark);

        return demandedMarkResponseTo;
    }

    public async Task<MarkResponseTo> UpdateMark(MarkRequestTo
updateMarkRequestTo)
    {
        Mark markFromDto = _mapper.Map<Mark>(updateMarkRequestTo);

        Mark updateMark =
            await _markRepository.UpdateAsync(markFromDto)
            ?? throw new MarkNotFoundException(
                $"Update mark {markFromDto} was not found");

        MarkResponseTo updateMarkResponseTo =
            _mapper.Map<MarkResponseTo>(updateMark);

        return updateMarkResponseTo;
    }
}

using AutoMapper;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
```

```
using rest1.application.exceptions;
using rest1.application.interfaces;
using rest1.application.interfaces.services;
using rest1.core.entities;

namespace rest1.application.services;

 public class NewsService : INewsService
    {
        private readonly IMapper _mapper;

        private readonly INewsRepository _newsRepository;

        public NewsService(IMapper mapper, INewsRepository repository)
        {
            _mapper = mapper;
            _newsRepository = repository;
        }

        public async Task<NewsResponseTo> CreateNews(NewsRequestTo
createNewsRequestTo)
        {
            News newsFromDto = _mapper.Map<News>(createNewsRequestTo);

            News createdNews = await _newsRepository.AddAsync(newsFromDto);

            NewsResponseTo dtoFromCreatedNews =
_mapper.Map<NewsResponseTo>(createdNews);

            return dtoFromCreatedNews;
        }

        public async Task DeleteNews(NewsRequestTo deleteNewsRequestTo)
        {
            News newsFromDto = _mapper.Map<News>(deleteNewsRequestTo);
            _ = await _newsRepository.DeleteAsync(newsFromDto) ?? throw new
NewNotFoundException($"Deleted news {newsFromDto} was not found");
        }

        public async Task<IEnumerable<NewsResponseTo>> GetAllNews()
```

```
        {
            IEnumerable<News> allNews = await _newsRepository.GetAllAsync();

            var allNewsResponseTos = new List<NewsResponseTo>();
            foreach (News news in allNews)
            {
                NewsResponseTo newsTo = _mapper.Map<NewsResponseTo>(news);
                allNewsResponseTos.Add(newsTo);
            }

            return allNewsResponseTos;
        }

        public async Task<NewsResponseTo> GetNews(NewsRequestTo
getNewsRequestTo)
        {
            News newsFromDto = _mapper.Map<News>(getNewsRequestTo);

            News demandedNews = await
_newsRepository.GetByIdAsync(newsFromDto.Id) ?? throw new
NewNotFoundException($"Demanded news {newsFromDto} was not found");

            NewsResponseTo demandedNewsResponseTo =
_mapper.Map<NewsResponseTo>(demandedNews);

            return demandedNewsResponseTo;
        }

        public async Task<NewsResponseTo> UpdateNews(NewsRequestTo
updateNewsRequestTo)
        {
            News newsFromDto = _mapper.Map<News>(updateNewsRequestTo);

            News? updateNews = await _newsRepository.UpdateAsync(newsFromDto)
?? throw new NewNotFoundException($"Update news {newsFromDto} was not
found");

            NewsResponseTo updateNewsResponseTo =
_mapper.Map<NewsResponseTo>(updateNews);
```

```
            return updateNewsResponseTo;
        }
    }

using AutoMapper;
using rest1.application.DTOs.requests;
using rest1.application.DTOs.responses;
using rest1.application.exceptions;
using rest1.application.interfaces;
using rest1.application.interfaces.services;
using rest1.core.entities;

namespace rest1.application.services;

public class NoteService : INoteService
    {
        private readonly IMapper _mapper;
        private readonly INoteRepository _noteRepository;

        public NoteService(IMapper mapper, INoteRepository repository)
        {
            _mapper = mapper;
            _noteRepository = repository;
        }

        public async Task<NoteResponseTo> CreateNote(NoteRequestTo
createNoteRequestTo)
        {
            Note noteFromDto = _mapper.Map<Note>(createNoteRequestTo);

            Note createdNote = await _noteRepository.AddAsync(noteFromDto);

            NoteResponseTo dtoFromCreatedNote =
                _mapper.Map<NoteResponseTo>(createdNote);

            return dtoFromCreatedNote;
        }

        public async Task DeleteNote(NoteRequestTo deleteNoteRequestTo)
        {
```

```
            Note noteFromDto = _mapper.Map<Note>(deleteNoteRequestTo);

            _ = await _noteRepository.DeleteAsync(noteFromDto)
                ?? throw new NoteNotFoundException(
                    $"Delete note {noteFromDto} was not found");
        }

        public async Task<IEnumerable<NoteResponseTo>> GetAllNotes()
        {
            IEnumerable<Note> allNotes =
                await _noteRepository.GetAllAsync();

            var response = new List<NoteResponseTo>();

            foreach (Note note in allNotes)
            {
                response.Add(_mapper.Map<NoteResponseTo>(note));
            }

            return response;
        }

        public async Task<NoteResponseTo> GetNote(NoteRequestTo
getNoteRequestTo)
        {
            Note noteFromDto = _mapper.Map<Note>(getNoteRequestTo);

            Note demandedNote =
                await _noteRepository.GetByIdAsync(noteFromDto.Id)
                ?? throw new NoteNotFoundException(
                    $"Demanded note {noteFromDto} was not found");

            return _mapper.Map<NoteResponseTo>(demandedNote);
        }

        public async Task<NoteResponseTo> UpdateNote(NoteRequestTo
updateNoteRequestTo)
        {
            Note noteFromDto = _mapper.Map<Note>(updateNoteRequestTo);
```

```
        Note updatedNote =
            await _noteRepository.UpdateAsync(noteFromDto)
            ?? throw new NoteNotFoundException(
                $"Update note {noteFromDto} was not found");

        return _mapper.Map<NoteResponseTo>(updatedNote);
    }
}

using System.Linq.Expressions;
using rest1.application.interfaces;

namespace rest1.application.specifications;

public abstract class BaseSpecification<T> : ISpecification<T>
{
    public Expression<Func<T, bool>> Criteria { get; private set; }

    public List<Expression<Func<T, object>>> Includes { get; } = [];

    public Expression<Func<T, object>>? OrderBy { get; private set; }

    public Expression<Func<T, object>>? OrderByDescending { get; private set; }

    public int? Take { get; private set; }

    public int? Skip { get; private set; }

    public bool IsPagingEnabled { get; private set; }

    protected BaseSpecification(Expression<Func<T, bool>> criteria)
    {
        Criteria = criteria;
    }

    protected void AddInclude(Expression<Func<T, object>> includeExpression)
    {
        Includes.Add(includeExpression);
    }
```

```
    protected void ApplyPaging(int skip, int take)
    {
        Skip = skip;
        Take = take;
        IsPagingEnabled = true;
    }

    protected void ApplyOrderBy(Expression<Func<T, object>>
orderByExpression)
    {
        OrderBy = orderByExpression;
    }

    protected void ApplyOrderByDescending(Expression<Func<T, object>>
orderByDescendingExpression)
    {
        OrderByDescending = orderByDescendingExpression;
    }
}

namespace rest1.core.entities;

public class Creator(
    string login,
    string password,
    string firstname,
    string lastname) : Entity
{
    public string Login { get; set; } = login;

    public string Password { get; set; } = password;

    public string Firstname { get; set; } = firstname;

    public string Lastname { get; set; } = lastname;
}

namespace rest1.core.entities;

public abstract class Entity
```

```
{
   public long Id { get; set; }
}

namespace rest1.core.entities;

public class Mark(string name) : Entity
{
   public string Name { get; set; } = name;
   public long[] NewsIds { get; set; } = new long[0];
}

namespace rest1.core.entities;

public class News : Entity
{
   public long CreatorId { get; set; }

   public string Title { get; set; } = string.Empty;

   public string Content { get; set; } = string.Empty;

   public DateTime CreatedAt { get; set; }

   public DateTime Modified { get; set; }

   // lookup object
   public Creator? Creator { get; set; }
   public long[]? MarkIds { get; set; }
}

namespace rest1.core.entities;

public class Note : Entity
{
   public long NewsId { get; set; }

   public string Content { get; set; } = string.Empty;

   // lookup object
```

```
    public News? News { get; set; }


}

using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.infrastructure.persistence;

public class CreatorRepository : Repository<Creator>, ICreatorRepository
{

}

using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.infrastructure.persistence;

public class MarkRepository: Repository<Mark>, IMarkRepository
{

}

using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.infrastructure.persistence;

public class NewsRepository: Repository<News>, INewsRepository
{

}

using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.infrastructure.persistence;
```

```
public class NoteRepository: Repository<Note>, INoteRepository
{

}

using System.Collections.Concurrent;
using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.infrastructure.persistence;

public class Repository<TEntity> : IRepository<TEntity> where TEntity : Entity
{
   protected readonly ConcurrentDictionary<long, TEntity> _entities = new();

   private long _nextId = 1;

   private readonly Lock _lock = new();

   public virtual Task<TEntity?> FindAsync(ISpecification<TEntity>
specification, CancellationToken cancellationToken = default)
   {
      cancellationToken.ThrowIfCancellationRequested();

      lock (_lock)
      {
         IQueryable<TEntity> query = ApplySpecification(specification);
         return Task.FromResult(query.FirstOrDefault());
      }
   }

   public virtual Task<IEnumerable<TEntity>>
FindAllAsync(ISpecification<TEntity> specification, CancellationToken
cancellationToken = default)
   {
      cancellationToken.ThrowIfCancellationRequested();

      lock (_lock)
      {
         IQueryable<TEntity> query = ApplySpecification(specification);
```

```
            return Task.FromResult(query.ToList().AsEnumerable());
        }
    }

    public virtual Task<int> CountAsync(ISpecification<TEntity> specification,
CancellationToken cancellationToken = default)
    {
        cancellationToken.ThrowIfCancellationRequested();

        lock (_lock)
        {
            IQueryable<TEntity> query = ApplySpecification(specification);
            return Task.FromResult(query.Count());
        }
    }

    public virtual Task<bool> AnyAsync(ISpecification<TEntity> specification,
CancellationToken cancellationToken = default)
    {
        cancellationToken.ThrowIfCancellationRequested();

        lock (_lock)
        {
            IQueryable<TEntity> query = ApplySpecification(specification);
            return Task.FromResult(query.Any());
        }
    }


    // CRUD
    public virtual Task<TEntity> AddAsync(TEntity entity, CancellationToken
cancellationToken = default)
    {
        ArgumentNullException.ThrowIfNull(entity);
        cancellationToken.ThrowIfCancellationRequested();

        lock (_lock)
        {
            var id = entity.Id;
            if (id == 0)
```

```
            {
               id = _nextId;
            }

            if (_entities.TryAdd(_nextId, entity))
            {
               entity.Id = id;
               _nextId++;
               return Task.FromResult(entity);
            }

            throw new InvalidOperationException($"Entity with id {id} already
exists");
         }
      }

      public virtual Task<TEntity?> UpdateAsync(TEntity entity, CancellationToken
cancellationToken = default)
      {
         ArgumentNullException.ThrowIfNull(entity);
         cancellationToken.ThrowIfCancellationRequested();

         lock (_lock)
         {
            var id = entity.Id;

            if (_entities.TryGetValue(id, out TEntity? updateEntity))
            {
               _entities[id] = entity;
               updateEntity = _entities[id];
            }

            return Task.FromResult(updateEntity);
         }
      }

      public virtual Task<TEntity?> DeleteAsync(TEntity entity, CancellationToken
cancellationToken = default)
      {
         ArgumentNullException.ThrowIfNull(entity);
```

```
        cancellationToken.ThrowIfCancellationRequested();

        lock (_lock)
        {
            var id = entity.Id;
            _entities.TryRemove(id, out TEntity? deletedEntity);

            return Task.FromResult(deletedEntity);
        }
    }

    public virtual Task<bool> DeleteByIdAsync(long id, CancellationToken
cancellationToken = default)
    {
        cancellationToken.ThrowIfCancellationRequested();

        lock (_lock)
        {
            return Task.FromResult(_entities.TryRemove(id, out _));
        }
    }
    public virtual Task<TEntity?> GetByIdAsync(long id, CancellationToken
cancellationToken = default)
    {
        cancellationToken.ThrowIfCancellationRequested();

        lock (_lock)
        {
            return Task.FromResult(_entities.GetValueOrDefault(id));
        }
    }
    public virtual Task<IEnumerable<TEntity>> GetAllAsync(CancellationToken
cancellationToken = default)
    {
        cancellationToken.ThrowIfCancellationRequested();

        lock (_lock)
        {
            return Task.FromResult(_entities.Values.AsEnumerable());
        }
```

```
    }
    protected virtual IQueryable<TEntity>
ApplySpecification(ISpecification<TEntity> spec)
    {
        IQueryable<TEntity> query = _entities.Values.AsQueryable();

        if (spec.Criteria != null)
        {
            query = query.Where(spec.Criteria);
        }
        if (spec.OrderBy != null)
        {
            query = query.OrderBy(spec.OrderBy);
        }
        else if (spec.OrderByDescending != null)
        {
            query = query.OrderByDescending(spec.OrderByDescending);
        }

        if (spec.IsPagingEnabled)
        {
            query = query.Skip(spec.Skip ?? 0)
                        .Take(spec.Take ?? 10);
        }
        return query;
    }
    public void Clear()
    {
        lock (_lock)
        {
            _entities.Clear();
        }
    }
}


using Microsoft.EntityFrameworkCore;
using rest1.core.entities;
using rest1.persistence.db.configurations;
```

```
namespace rest1.persistence.db;

public class RestServiceDbContext : DbContext
{
   public RestServiceDbContext(DbContextOptions<RestServiceDbContext>
options)
       : base(options) { }

   public DbSet<Creator> Creators { get; set; }

   public DbSet<Note> Notes { get; set; }

   public DbSet<News> News { get; set; }

   public DbSet<Mark> Marks { get; set; }

   protected override void OnModelCreating(ModelBuilder modelBuilder)
   {
      modelBuilder.ApplyConfiguration(new CreatorConfiguration());
      modelBuilder.ApplyConfiguration(new MarkConfiguration());
      modelBuilder.ApplyConfiguration(new NoteConfiguration());
      modelBuilder.ApplyConfiguration(new NewsConfiguration());
      base.OnModelCreating(modelBuilder);
   }
}

using Microsoft.EntityFrameworkCore;
using Microsoft.EntityFrameworkCore.Metadata.Builders;
using rest1.core.entities;

namespace rest1.persistence.db.configurations;

public class CreatorConfiguration : IEntityTypeConfiguration<Creator>
{
   public void Configure(EntityTypeBuilder<Creator> builder)
   {
      builder.ToTable("tbl_creator", schema: "public");
      builder.Property(e => e.Id).HasColumnName("id");
      builder.Property(e => e.Login).HasColumnName("login");
      builder.Property(e => e.Password).HasColumnName("password");
```

```
        builder.Property(e => e.Firstname).HasColumnName("firstname");
        builder.Property(e => e.Lastname).HasColumnName("lastname");
    }
}

using Microsoft.EntityFrameworkCore;
using Microsoft.EntityFrameworkCore.Metadata.Builders;
using rest1.core.entities;

namespace rest1.persistence.db.configurations;

public class MarkConfiguration : IEntityTypeConfiguration<Mark>
{
    public void Configure(EntityTypeBuilder<Mark> builder)
    {
        builder.ToTable("tbl_mark", schema: "public");
        builder.Property(m => m.Id).HasColumnName("id");
        builder.Property(m => m.Name).HasColumnName("name");
    }
}

using Microsoft.EntityFrameworkCore;
using Microsoft.EntityFrameworkCore.Metadata.Builders;
using rest1.core.entities;

namespace rest1.persistence.db.configurations;

public class NewsConfiguration : IEntityTypeConfiguration<News>
{
    public void Configure(EntityTypeBuilder<News> builder)
    {
        builder.ToTable("tbl_news", schema: "public");
        builder.Property(n => n.Id).HasColumnName("id");
        builder.Property(n => n.CreatorId).HasColumnName("creator_id");
        builder.Property(n => n.Title).HasColumnName("title");
        builder.Property(n => n.Content).HasColumnName("content");
        builder.Property(n => n.CreatedAt).HasColumnName("created");
        builder.Property(n => n.Modified).HasColumnName("modified");
        builder.HasMany(n => n.Marks)
            .WithMany(m => m.News)
```

```
            .UsingEntity<Dictionary<string, object>>("tbl_newsMark",
                j => j.HasOne<Mark>()
                    .WithMany()
                    .HasForeignKey("MarkId")
                    .HasConstraintName("FK_MarkNews_Mark"),
                j => j.HasOne<News>()
                    .WithMany()
                    .HasForeignKey("NewsId")
                    .HasConstraintName("FK_MarkNews_News")
                    .OnDelete(DeleteBehavior.Cascade),
                j => j.HasKey("MarkId", "NewsId")
            );
    }
}

using Microsoft.EntityFrameworkCore;
using Microsoft.EntityFrameworkCore.Metadata.Builders;
using rest1.core.entities;

namespace rest1.persistence.db.configurations;

public class NoteConfiguration : IEntityTypeConfiguration<Note>
{
    public void Configure(EntityTypeBuilder<Note> builder)
    {
        builder.ToTable("tbl_note", schema: "public");
        builder.Property(p => p.Id).HasColumnName("id");
        builder.Property(p => p.NewsId).HasColumnName("news_id");
        builder.Property(p => p.Content).HasColumnName("content");
    }
}

using Microsoft.EntityFrameworkCore;
using rest1.application.exceptions.db;
using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.persistence.db.repositories;

public class DbCreatorRepository : DbRepository<Creator>, ICreatorRepository
```

```
{
   public DbCreatorRepository(RestServiceDbContext dbContext) :
base(dbContext)
   {
   }

   public override async Task<Creator> AddAsync(Creator entity,
CancellationToken cancellationToken = default)
   {
      ArgumentNullException.ThrowIfNull(entity);
      cancellationToken.ThrowIfCancellationRequested();

      var existingEntity = await _dbSet.Where(e => e.Id == entity.Id || e.Login ==
entity.Login).FirstOrDefaultAsync(cancellationToken);

      if (existingEntity != null)
      {
         throw new InvalidOperationException($"Entity with id {entity.Id} already
exists");
      }

      var entry = await _dbSet.AddAsync(entity, cancellationToken);
      try
      {
         await _dbContext.SaveChangesAsync(cancellationToken);
      }
      catch (DbUpdateException ex) when (IsForeignKeyViolation(ex))
      {
         throw new ForeignKeyViolationException("Foreign key doesn't exist");
      }

      return entry.Entity;
   }
}

using Microsoft.EntityFrameworkCore;
using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.persistence.db.repositories;
```

```
public class DbMarkRepository : DbRepository<Mark>, IMarkRepository
{
   public DbMarkRepository(RestServiceDbContext dbContext) : base(dbContext)
   {
   }

   public async Task DeleteMarksWithoutNews()
   {
      var emptyMarkers = await _dbSet.Where(m => m.News.Count ==
0).ToListAsync();
      _dbSet.RemoveRange(emptyMarkers);
      await _dbContext.SaveChangesAsync();
   }
}

using Microsoft.EntityFrameworkCore;
using rest1.application.exceptions.db;
using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.persistence.db.repositories;

 public class DbNewsRepository : DbRepository<News>, INewsRepository
{
   public DbNewsRepository(RestServiceDbContext dbContext) : base(dbContext)
   {

   }

   public override async Task<News> AddAsync(News entity, CancellationToken
cancellationToken = default)
   {

      ArgumentNullException.ThrowIfNull(entity);
      cancellationToken.ThrowIfCancellationRequested();

      var existingEntity = await _dbSet.Where(n => n.Id == entity.Id || n.Title ==
entity.Title).FirstOrDefaultAsync(cancellationToken);
```

```
        if (existingEntity != null)
        {
            throw new InvalidOperationException($"Entity with id {entity.Id} already
exists");
        }

        var entry = await _dbSet.AddAsync(entity, cancellationToken);
        try
        {
            await _dbContext.SaveChangesAsync(cancellationToken);
        }
        catch (DbUpdateException ex) when (IsForeignKeyViolation(ex))
        {
            throw new ForeignKeyViolationException("Foreign key doesn't exist");
        }

        return entry.Entity;
    }

    public async override Task<News?> DeleteAsync(News entity,
CancellationToken cancellationToken = default)
    {
        ArgumentNullException.ThrowIfNull(entity);
        cancellationToken.ThrowIfCancellationRequested();

        var existingEntity = await _dbSet.Where(n => n.Id == entity.Id)
            .Include(n => n.Marks)
            .FirstOrDefaultAsync();

        if (existingEntity == null)
            return null;

        _dbSet.Remove(existingEntity);

        await _dbContext.SaveChangesAsync(cancellationToken);

        return existingEntity;
    }

    public async Task<News> GetByIdWithMarksAsync(int id)
```

```
    {
        return await _dbSet
            .Include(n => n.Marks)
            .FirstOrDefaultAsync(n => n.Id == id);
    }

    public async Task RemoveAsync(News news)
    {
        _dbSet.Remove(news);
        await _dbContext.SaveChangesAsync();
    }
}

using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.persistence.db.repositories;

public class DbNoteRepository : DbRepository<Note>, INoteRepository
{
    public DbNoteRepository(RestServiceDbContext dbContext) : base(dbContext)
    {
    }
}

using Microsoft.EntityFrameworkCore;
using Npgsql;
using rest1.application.exceptions.db;
using rest1.application.interfaces;
using rest1.core.entities;

namespace rest1.persistence.db.repositories;


 public class DbRepository<TEntity> : IRepository<TEntity>
        where TEntity : Entity
    {
        protected readonly DbSet<TEntity> _dbSet;
        protected readonly DbContext _dbContext;
```

```
        public DbRepository(RestServiceDbContext dbContext)
        {
            _dbContext = dbContext ?? throw new
ArgumentNullException(nameof(dbContext));
            _dbSet = dbContext.Set<TEntity>();
        }

        public virtual async Task<TEntity?> FindAsync(ISpecification<TEntity>
specification, CancellationToken cancellationToken = default)
        {
            cancellationToken.ThrowIfCancellationRequested();

            IQueryable<TEntity> query = ApplySpecification(specification);
            return await query.FirstOrDefaultAsync(cancellationToken);
        }

        public virtual async Task<IEnumerable<TEntity>>
FindAllAsync(ISpecification<TEntity> specification, CancellationToken
cancellationToken = default)
        {
            cancellationToken.ThrowIfCancellationRequested();

            IQueryable<TEntity> query = ApplySpecification(specification);
            return await query.ToListAsync(cancellationToken);
        }

        public virtual async Task<int> CountAsync(ISpecification<TEntity>
specification, CancellationToken cancellationToken = default)
        {
            cancellationToken.ThrowIfCancellationRequested();

            IQueryable<TEntity> query = ApplySpecification(specification);
            return await query.CountAsync(cancellationToken);
        }

        public virtual async Task<bool> AnyAsync(ISpecification<TEntity>
specification, CancellationToken cancellationToken = default)
        {
            cancellationToken.ThrowIfCancellationRequested();
```

```
        IQueryable<TEntity> query = ApplySpecification(specification);
        return await query.AnyAsync(cancellationToken);
    }

    // CRUD
    public virtual async Task<TEntity> AddAsync(TEntity entity,
CancellationToken cancellationToken = default)
    {
        ArgumentNullException.ThrowIfNull(entity);
        cancellationToken.ThrowIfCancellationRequested();

        var existingEntity = await _dbSet.FindAsync([entity.Id],
cancellationToken: cancellationToken);

        if (existingEntity != null)
        {
            throw new InvalidOperationException($"Entity with id {entity.Id}
already exists");
        }

        var entry = await _dbSet.AddAsync(entity, cancellationToken);
        try
        {
            await _dbContext.SaveChangesAsync(cancellationToken);
        }
        catch (DbUpdateException ex) when (IsForeignKeyViolation(ex))
        {
            throw new ForeignKeyViolationException("Foreign key doesn't exist");
        }

        return entry.Entity;
    }

    protected static bool IsForeignKeyViolation(DbUpdateException ex)
    {
        return ex.InnerException switch
        {
            PostgresException pgEx => pgEx.SqlState == "23503",
            _ => false
        };
```

```
        }

        public virtual async Task<TEntity?> UpdateAsync(TEntity entity,
CancellationToken cancellationToken = default)
        {
            ArgumentNullException.ThrowIfNull(entity);
            cancellationToken.ThrowIfCancellationRequested();

            var existingEntity = await _dbSet.FindAsync([entity.Id,
cancellationToken], cancellationToken: cancellationToken);

            if (existingEntity == null)
                return null;

            _dbContext.Entry(existingEntity).CurrentValues.SetValues(entity);
            await _dbContext.SaveChangesAsync(cancellationToken);

            return existingEntity;
        }

        public virtual async Task<TEntity?> DeleteAsync(TEntity entity,
CancellationToken cancellationToken = default)
        {
            ArgumentNullException.ThrowIfNull(entity);
            cancellationToken.ThrowIfCancellationRequested();

            var existingEntity = await _dbSet.FindAsync(entity.Id, cancellationToken);

            if (existingEntity == null)
                return null;

            _dbSet.Remove(existingEntity);
            await _dbContext.SaveChangesAsync(cancellationToken);

            return existingEntity;
        }

        public virtual async Task<bool> DeleteByIdAsync(long id,
CancellationToken cancellationToken = default)
        {
```

```
        cancellationToken.ThrowIfCancellationRequested();

        var entity = await _dbSet.FindAsync(id , cancellationToken);

        if (entity == null)
            return false;

        _dbSet.Remove(entity);
        await _dbContext.SaveChangesAsync(cancellationToken);

        return true;
    }

    public virtual async Task<TEntity?> GetByIdAsync(long id,
CancellationToken cancellationToken = default)
    {
        cancellationToken.ThrowIfCancellationRequested();

        return await _dbSet.FindAsync(id, cancellationToken);
    }

    public virtual async Task<IEnumerable<TEntity>>
GetAllAsync(CancellationToken cancellationToken = default)
    {
        cancellationToken.ThrowIfCancellationRequested();

        return await _dbSet.ToListAsync(cancellationToken);
    }

    protected virtual IQueryable<TEntity>
ApplySpecification(ISpecification<TEntity> spec)
    {
        IQueryable<TEntity> query = _dbSet.AsQueryable();

        if (spec.Criteria != null)
        {
            query = query.Where(spec.Criteria);
        }

        // Применяем инклюды (включаемые навигационные свойства)
```

```
        query = spec.Includes
            .Aggregate(query, (current, include) => current.Include(include));

        if (spec.OrderBy != null)
        {
            query = query.OrderBy(spec.OrderBy);
        }
        else if (spec.OrderByDescending != null)
        {
            query = query.OrderByDescending(spec.OrderByDescending);
        }

        if (spec.IsPagingEnabled)
        {
            query = query.Skip(spec.Skip ?? 0)
                         .Take(spec.Take ?? 10);
        }

        return query;
    }
}
```